Code No: 234AA **KR23**

KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech II Year II Semester Regular Examinations, June/July-2025

AUTOMATA THEORY AND COMPILER DESIGN

(COMMON TO IT, CSE (AI&ML), CSE(DS))

Time: 3 hours Max. Marks: 60

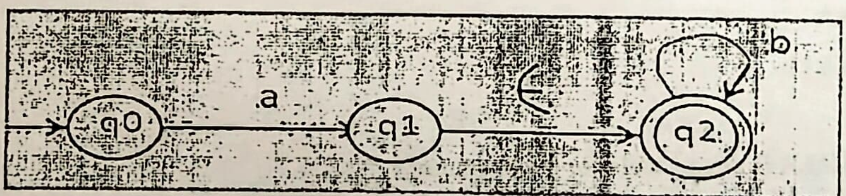
Note: This question paper contains two parts A and B.
Part A is compulsory which carries 10 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b as sub questions.

PART- A (10 Marks)

1. a) List types of Automata. [1M] L1
   b) Define NFA. [1M] L1
   c) List the types of grammar. [1M] L1
   d) List types of derivation. [1M] L1
   e) What are the components of Turing machine? [1M] L1
   f) Explain two buffer scheme. [1M] L2
   g) What is syntax directed translation? [1M] L1
   h) Explain quadruple. [1M] L2
   i) Define activation record. [1M] L1
   j) State use of loop jamming. [1M] L1

PART- B (5 X 10 =50 Marks)

2. Convert NFA with E moves to NFA without E moves. [10M] L3



OR

3. a) Construct DFA which accepts password of length 6 over Z= {u, b}. Specify logic and draw transition diagram and transition table. [5M] L3

b) Construct DFA to represent working of switch. The switch takes two inputs 0 and 1. If input is 0 then switch is OFF. It input is 1 then switch is ON. Specify logic and draw transition diagram and transition table. [5M] L3

4. a) Differentiate between regular grammar and context free grammar [5M] L4

b) Define ambiguous grammar. Identify whether the given grammar is ambiguous or not. [5M] L3

S -> ABA

A -> Aa | ε

B -> Bb | ε

OR

5. Construct PDA for language $L = \{a^n b^n \mid n >= 1\}$, Write logic, transition table and draw transition diagram. [10M] L3

6. Construct Turing machine for $L=\{ a^n b^n c^n \mid n >= 1\}$ [10M] L3

OR

7. a) With a neat diagram show the phases of compiler.will process the program assignment Statement for position =initial + rate*60 [5M] L3

b) Explain the lexical analysis and types of buffering in detail [5M] L3

8. Construct CLR (1) parse table for given grammar. [10M] L3

S -> AA

A -> aA | b

OR

9. Construct LALR (1) parse table for given grammar. [10M] L3

S -> AA

A -> aA | b

10. Explain simple code generation algorithm. Give suitable example. [10M] L2

OR

11. Explain different storage allocations strategies in detail. [10M] L2

---ooOoo---

Code No: 234AA **KR23**

KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

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B.Tech II Year II Semester Regular Examinations, June/July-2025

AUTOMATA THEORY AND COMPILER DESIGN

(COMPUTER SCIENCE AND ENGINEERING)

Time: 3 hours Max. Marks: 60

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 10 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b as sub questions.

PART- A

(10 Marks)

1. a) Define Finite Automata. [1M] L1
   b) What is pumping lemma? [1M] L1
   c) Define Ambiguous grammar. [1M] L1
   d) What are the components of Push down Automata? [1M] L1
   e) List the phases of compiler. [1M] L1
   f) Explain transition diagram of Turing machine. [1M] L2
   g) What is recursive descent parser [1M] L1
   h) List the representation of three address code [1M] L1
   i) What is Directed Acyclic Graph? [1M] L1
   j) List storage allocation strategies. [1M] L1

PART- B

(5 X 10 =50 Marks)

2. a) Convert given NFA transition table below to DFA. [5M] L3

δ	0	1
->A	A	{A, B}
B	Φ	C
*c	Φ	Φ

b) Discuss Pumping lemma. Prove that language $L = \{a^n b^n \mid n >= 1\}$ is not regular. [5M] L3

OR

3. Construct DFA for language which accepts strings over an alphabet Z {0, 1}. [10M] L3

a. Where length of string =2.

b. String starts with ' 01'

Specify logic and draw transition diagram and transition table.

4. Explain Chomsky Hierarchy? Give an example. [10M] L2

OR

5. Explain structural representation of Push Down Automata with suitable example [10M] L3

6. Define Turning machine and explain its structural representation [10M] L2

OR

7. Explain in detail Phases of compiler with example [10M] L2

8. a) Explain types of parsing with example. Construct recursive decent parser for given grammar: [5M] L3

$E \rightarrow iE'$

$E' \rightarrow +IE' \mid \varepsilon$

b) Construct LL (I) parser for given grammar [5M] L3

$S \rightarrow aABb$

$A \rightarrow c \mid \varepsilon$

$B \rightarrow d \mid \varepsilon$

OR

9. Construct Quadruple, Triple, Indirect triple for expression [10M] L3

$t = p^* - q + p^* - q$

10. a) Explain code optimization techniques with example. [5M] L2

b) Explain Directed Acyclic Graph, its applications with example. [5M] L2

OR

11. Explain issues in code generation with example. [10M] L2

---ooOoo---

Code No:224AE

RKR21

KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

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Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech II Year II Semester Supplementary Examinations, June/July -2025

AUTOMATA THEORY AND COMPILER DESIGN

(COMMON TO CSE, IT, CSE(AI&ML), CSE(DS))

Time: 3 hours Max. Marks: 70

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 20 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b as sub questions.

PART- A (20 Marks)

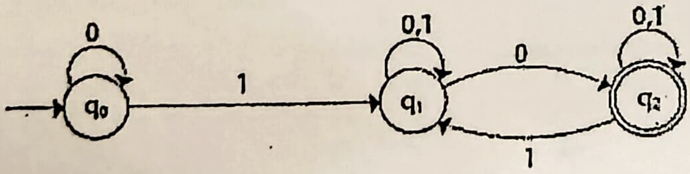
1. a) Write the differences between NFA and DFA. [2M] L1
   b) Draw a DFA which accepts any no of a's with $\Sigma=\{a\}$. [2M] L2
   c) Define Ambiguous Grammar. [2M] L1
   d) Define PDA. [2M] L1
   e) Write the role of compiler in compilation process. [2M] L1
   f) List the differences between top down parsing and bottom up parsing. [2M] L1
   g) Define SDT. [2M] L1
   h) List out the differences between S-attribute and L-attribute grammar. [2M] L1
   i) Define Activation record. [2M] L1
   j) What is a basic block with an example? [2M] L1

PART- B (5 X 10 =50 Marks)

2. a) Construct NFA to accept all strings of a's and b's which contains substring as aa or bb. [5M] L3
   b) Construct Finite Automata for the regular Expression 1(01+10)*00 [5M] L3

OR

3. Convert the given NFA to equivalent DFA [10M] L2



4. Consider the CFG with the following production rules:

$S \rightarrow aB \mid bA \quad A \rightarrow bAA \mid aS \mid a \quad B \rightarrow aBB \mid bS \mid b$

Give the LMD,RMD and draw derivation tree for the string **abbaab** [10M] L2

OR

5. a) What is left recursion? How to eliminate it with an example grammar. [5M] L2

b) Construct PDA for the following Language $L=\{w\#w^R \mid \#$ is special symbol and $w \in (0+1)^*\}$? [5M] L3

6. a) Construct Turing Machine for the language $L=\{a^nb^nc^n \mid n>0\}$ [5M] L3

b) Construct the CLR Parsing table for following grammar [5M] L3

$S \rightarrow CC$

$C \rightarrow aC \mid d$

OR

7. Explain the different phases of a compiler, showing the output of each phase, using the following statement:
Position: =initial+ rate*60 where position, initial, rate are real type. [10M] L3

8. Define Syntax directed definition? Give SDD to evaluate arithmetic expression and show annotated parse tree that evaluate the expression 2*3+4? [10M]L3

OR

9. Explain three address code representation. Build the quadruple, triple forms of three address code

Statements $a = b + c * d$ [10M] L3

10. Explain about the various storage allocation methods with examples. [10M] L2

OR

11.a) Define optimization. What are the different techniques of loop optimization? [5M]L2

b) Write the different issues in code generation algorithm. [5M] L2

---ooOoo---

Code No:214AE **KR21**

KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

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Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.TECH II YEAR II SEMESTER REGULAR EXAMINATIONS, AUGUST - 2023

AUTOMATA THEORY AND COMPILER DESIGN

(COMPUTER SCIENCE AND ENGINEERING)

Time: 3 hours **Max. Marks: 70**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 20 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b, c as sub questions.

PART- A (2 X 10 =20 Marks)

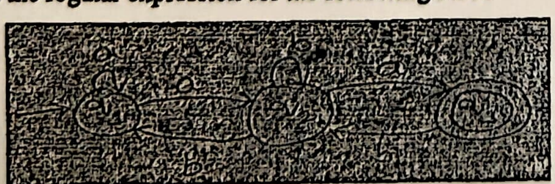
1. a) Define ϵ - NFA with Example. [2M]
   b) List any two algebraic properties of Regular Expressions [2M]
   c) What is an ambiguity [2M]
   d) Compare DPDA and NPDA. [2M]
   e) Define Turing Machine [2M]
   f) Differentiate Compiler vs Interpreter [2M]
   g) Define an attribute. Give the types of attributes [2M]
   h) List out the types of three address code. [2M]
   i) Define DAG. [2M]
   j) Define Stack. [2M]

PART- B (5 X 10 =50 Marks)

2. a) Define DFA. Construct a minimal DFA over {a,b} where language L={$a^n b^m$ | n,m>=1} [5M]
   b) Construct the DFA where [5M]
      a. Strings starts with 'a'
      b. Strings end with 'a'
      c. Strings containing 'a'.
      on ∑= (a,b)

OR

3. Find the regular expression for the following NFA [10M]



P.T.O

Code No:154AD

KR20

KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech II Year II Semester Supplementary Examinations, June/July-2025

AUTOMATA AND COMPILER DESIGN

(COMMON TO CSE, IT, CSE(AI&ML), CSE(DS))

Time: 3 hours Max. Marks: 75

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b as sub questions.

PART- A (25 Marks)

1. a) Define Finite Automa. [2M] L1
   b) Give differences between DFA and NFA. [3M] L2
   c) Define Regular Expression with an example. [2M] L1
   d) Write the Language for the regular expression (0+1)*. [3M] L2
   e) Define Context Free Grammar with example. [2M] L1
   f) Give Differences between Bottom up Parsing & Top down parsing. [3M] L2
   g) What is Syntax Directed translation ? [2M] L1
   h) Define types of Intermediate Codes. [3M] L2
   i) Define Flow graph with an example. [2M] L1
   j) Explain Steps in Register Allocation & Assignment. [3M] L2

PART- B (5 X 10 =50 Marks)

2. Convert the following ε-NFA to NFA. [10M] L3

δ	0	1	ε
→A	{B, C}	A	B
B	ϕ	B	C
*C	C	C	ϕ

OR

3. Design an NFA & DFA in neat steps with $\sum = \{0, 1\}$ in which double '1' is followed by double '0' along with the transition table and write the intital and final stats in detail. [10 M] L4

4. Explain about the role of Lexical Analyzer. [10 M] L3

OR

5. a) Explain about the Structure of Compiler. [5M] L3

b) Define Left Most Derivation(LMD) and Right Most Derivation(RMD) . [5M] L4

Find the LMD and RMD from the following grammar

S -> baAaS | ab

A -> Aab | aa

For the string w =baaaaababaab

6. a) Describe about Predicting Parsing in detail. [5M] L3

b) Explain about shift Reducing Parsing and its Classification with an example for each. [5M] L4

OR

7. a) Distinguish between LR , CLR , LALR Parsers. [5M] L3

b) How to Examine the String Acceptance in LR-Parsing ? [5M] L4

8. a) Write in detail about evaluation order for SDD's . [5M] L3

b) Describe about Syntax Directed Transition Schemes. [5M] L2

OR

9. a) Explain about Stack Allocation in Run Time Environments. [5M] L3

b) Explain in detail about Implementation of Activation Record. [5M] L2

10. a) Explain about types of Optimization Techniques. [5M] L3

b) Explain about Basic Blocks and Construction of Flowgraphs. [5M] L2

OR

11. a) Explain about issues in design of a Flow Graph. [5M] L3

b) Explain Principle Sources of Optimization. [5M] L2

---ooOoo---

Code No:154AD

KR20

KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

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Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech II Year II Semester Supplementary Examinations, February-2024

AUTOMATA AND COMPILER DESIGN

(COMMON TO CSE, IT, CSE (AI&ML) & CSE (DATASCIENCE))

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b, c as sub questions.

PART- A (25 Marks)

1. a) Define Alphabet. [2M]L1
   b) Compare NFA & DFA. [3M]L2
   c) What is the role of Scanner? [2M]L1
   d) Specify the functionality of compiler. [3M]L2
   e) What is a handle in bottom-up parsing? Give an example. [2M]L2
   f) Describe the types of LR parsers. [3M]L2
   g) What is dynamic storage allocation? [2M]L1
   h) What is type checking? [3M]L2
   i) Define Peephole Optimization. [2M]L2
   j) Specify the applications of DAG. [3M]L2

PART- B (5 X 10 =50 Marks)

2. How to convert a regular expression to NFA? Explain with an example. [10M]L2

OR

3. Construct DFA equivalent to the NFA { { P, Q, R, S }, {0, 1 }, , P, {P, Q} } , where the transition function given by following table: [10M]L3

States	0	1
P	Q, S	Q
Q	R	Q, R
R	S	S
S	Q	P

4. Define Ambiguous grammar? Explain with an Example. [10M]L2

OR

5. a) Define Derivation tree. Explain about LMD and RMD. [5M]L3
   b) Considering the following grammar, remove left recursion and left factor. [5M]L3

$L \rightarrow L ; S \mid S$
$S \rightarrow id = E \mid id\ (E)$
$E \rightarrow id \mid num$

6. Define an LL(1) grammar. Is the following grammar LL(1). [10M]L3

$S \rightarrow iEtS \mid iEtSes \mid a$,
$E \rightarrow b$.

Also write the rules for computing FIRST() and FOLLOW().

OR

7. Discuss the construction of LR parser. What are the various data structures used in LR parser design? Discuss the construction of ACTION[ ] and GOTO[ ] table. [10M]L3

8. Explain syntax directed definitions. Define syntax directed definition for simple desk calculator. [10M]L2

OR

9. Discuss about the Stack allocation strategy of Run time environment with an example. [10M]L3

10. a) Explain the code generation algorithm and generate code for the following Expression:
$X = (a - b) + (a + c)$ [5M]L3

b) Define code optimization. Explain in detail about the Principal sources of optimization and optimization techniques. [5M]L2

OR

11. a) Explain in brief about the issues in the design of a code generator. [5M]L2

b) What is a basic block? With suitable example discuss various transformations on the basic block. [5M]L2

---ooOoo---

Code No: 214AE

KR21

KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.TECH II YEAR II SEMESTER REGULAR EXAMINATIONS, AUGUST - 2023

AUTOMATA THEORY AND COMPILER DESIGN

(COMMON TO IT, CSE (DATASCIENCE) & CSE (AI&ML))

Time: 3 hours **Max. Marks: 70**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 20 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b, c as sub questions.

PART- A (2 X 10 =20 Marks)

1. a) Define the terms symbol, string and Language. [2M]
   b) Define Regular Expression. [2M]
   c) Define Grammar? [2M]
   d) Discuss the formal definition of PDA [2M]
   e) What do you mean by Instantaneous Description of Turing Machine? [2M]
   f) Define Compiler? [2M]
   g) List out the types of LR parsers. [2M]
   h) Define SDT [2M]
   i) Define Heap. [2M]
   j) List out types of intermediate codes? [2M]

PART- B (5 X 10 =50 Marks)

2. a) Distinguish DFA and NFA. [5M]
   b) Design DFA for the following over {a, b} [5M]
      i) All strings that has atleast length is 2
      ii) All strings that has atmost length is 2
      iii) All strings that has exactly length is 2

OR

3. a) Find out the regular expression where Σ={a,b} [5M]
      a) Where length is at least 2
      b) Where length is even
      c) Where length is odd
   b) Convert Regular Expression (11+0)*(00+1)* to Finite Automata [5M]

P.T.O

4. a) Prove that the given language $L=\{0^n 1^n \mid n>=1\}$ is not a regular [5M]

b) Explain left most and right most derivations with examples. [5M]

OR

5. Construct a PDA for the given language $L = \{a^n b^n / n>=1\}$ and write the instantaneous description for the string "aaabbb". [10M]

6. Design a Turing machine for $L = \{ a^n b^n \mid n \geq 1\}$ [10M]

OR

7. Explain about the phases of Compiler. [10M]

8. a) Differentiate between Top down and Bottom up parsing techniques. [5M]

b) Construct SLR parsing table for the following grammar [5M]

E->E+T|T
T->T*F|F
F->(E)/id

OR Kmit

9. Construct an LALR Parsing table for the following grammar: [10M]

S→ aAb/e
A→ Bd
B→ g

10. Define Code Optimization. Explain about the types of Code Optimization. [10M]

OR

11. Explain in the DAG representation of the basic block with example. [10M]

--ooOoo--

Code No: 154AD **KR20**

KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech II Year II Semester Supplementary Examinations, September-2023

AUTOMATA AND COMPILER DESIGN

(COMMON TO CSE, IT, CSE (AI&ML) & CSE (DATASCIENCE))

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b, c as sub questions.

PART- A (25 Marks)

1. a) Define DFA with example. [2M]

b) List out the Applications of Regular Expressions. [3M]

c) Define Token, Pattern and Lexeme. [2M]

d) What is Context free grammar? [3M]

e) Distinguish top down and bottom-up parsing. [2M]

f) Discuss clearly about YACC. [3M]

g) Demonstrate the Three-Address Code. [2M]

h) Explain about L-Attributed SDD's. [3M]

i) Classify the Types of optimization techniques. [2M]

j) Construct the Peephole Optimization. [3M]

PART- B (5 X 10 =50 Marks)

2. a) Construct DFA equivalent to the NFA ({p, q, r, s}, {0,1}, δ, p, {s}) where the transition function δ is given in following table: [6M]

States	0	1
p	p, q	p
q	r	r
r	s	-
s	s	s

b) Show that $L=\{a^{2n}/n<0\}$ is Regular. [4M]

OR

3. a) Extend the Algebraic Laws for Regular Expressions. [5M]

b) Contrast the Construction of Finite Automata from Regular Expression. [5M]

4. a) Analyze about the elimination of Ambiguity. [5M]

b) Build a Derivation Tree with example. [5M]

P

5. a) Define the Grammar And Explain about Types of Grammars. [5M]

b) Conclude the Input Buffering and Tokens. [5M]

OR

6. a) Solve the following grammar [6M]

Lexp--> atom | list
Atom→ number | identifier
List→(lexp_seq)
lexp_seq→ lexp_seq lexp | lexp.
Remove left recursion and Construct LL(1) parsing table for the resulting grammar.

b) Inspect the Top-Down parsing with example. [4M]

OR

7. a) Construct SLR Parsing table for the grammar S→L=R/R, L→*R/id, R→L. [5M]

b) Examine the string acceptance in predictive parsing. [5M]

8. a) What is activation record? Discuss the structure of a typical activation record. What do you mean by a layout of a activation record? [5M]

b) Define and Discuss about the SDD's. [5M]

OR

Kmit

9. a) Simplify the sequence of three-address code instructions corresponding to the following: fragment using the above attribute grammar [6M]

for i= 1 to n step m+k do
s:= s+i

b) Organize the Overview of Storage Organization. [4M]

10. a) Define a Directed Acyclic Graph. Construct a DAG and write the sequence of instructions for the expression a + a * (b – c) + (b – c) * d . [6M]

b) State and explain different machine dependent code optimization techniques. [4M]

OR

11. a) Explain about the principle sources of Optimization. [5M]

b) Develop a machine code using generator Algorithm. [5M]

---ooOoo---